



Seasonal and Inventory-Driven Analysis of Washington Real Estate

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About Me

Background

Data analytics student with a focus on real estate market dynamics and statistical modeling.

Expertise

Python programming, multiple linear regression, and market trend analysis using real-world datasets.

Project Focus

Analyzing Washington State residential real estate to uncover data-driven insights for sellers, agents, and market participants.

Goal

Transform industry assumptions into statistical evidence through rigorous quantitative analysis.



THE PROBLEM

Research Question

Is there a statistically significant relationship between seasonal market patterns and inventory conditions on days-on-market and sale-to-list price ratios across Washington State residential real estate?

Why This Matters

Industry Practice

Real estate professionals base listing strategies on experience rather than statistical evidence.

Market Dynamics

Supply-and-demand conditions are sensitive to seasonality and inventory levels.

Data-Driven Framework

Quantitative modeling provides objective insights for market timing decisions.

Hypotheses

1

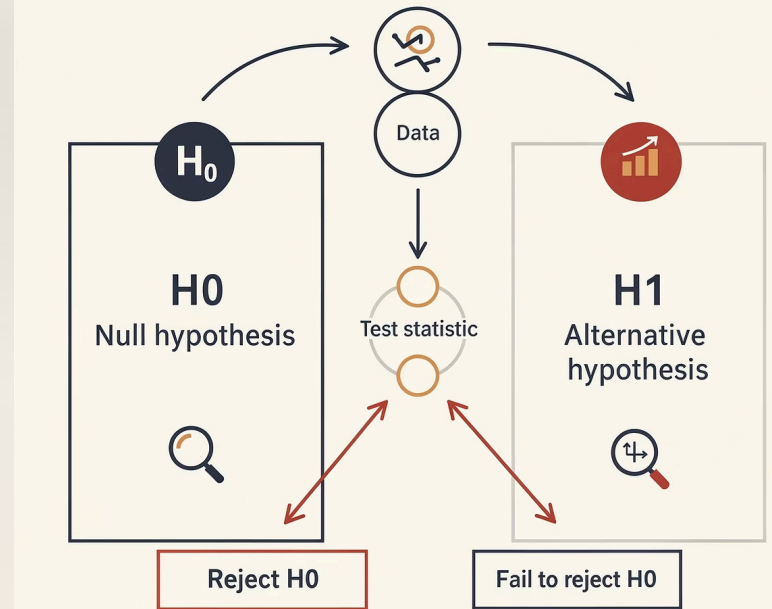
Null Hypothesis

Seasonality and inventory conditions have no statistically significant impact on real estate outcomes.

2

Alternative Hypothesis

Seasonal timing and inventory conditions significantly affect days on market and sale-to-list price ratios (adjusted $R^2 \geq 0.10$).



DATA COLLECTION

Dataset Overview

8,144

Total Rows

Monthly observations before cleaning

2012–2025

Time Period

Years of market data analyzed

20

Variables

Columns tracking market metrics

Data sourced from Redfin Data Center, covering Washington State county and metro-level aggregated residential real estate metrics.

Data Source Strengths

REDFIN

- Aggregated data reduces noise from individual property characteristics
- Public availability enhances reproducibility
- Redfin's industry reputation ensures data reliability
- Consistent monthly observations enable trend identification

Data Cleaning Process

01

Import & Validation

Loaded dataset using Python Pandas library and validated structure (8,143 rows × 20 columns).

03

Feature Engineering

Created seasonal categories, months-of-inventory variable, and time-based features.

02

Fix Missing Values

Applied forward fill to 8,093 missing region values due to formatting conventions.

04

Outlier Management

Applied 95th percentile capping to days on market and months of inventory.

Key Variables Created



Seasonal Categories

Mapped months to spring, summer, fall, and winter for trend analysis.



Months of Inventory

Calculated as inventory divided by homes sold—a key supply-demand indicator.



Time Features

Extracted year, month, and quarter for temporal analysis.

```
df["Homes Sold"] = pd.to_numeric(df["Homes Sold"], errors="coerce")
df["Inventory"] = pd.to_numeric(df["Inventory"], errors="coerce")

# Months of Inventory = Inventory / Homes Sold
df["Months_Inventory"] = np.where(df["Homes Sold"] > 0, df["Inventory"] / df["Homes Sold"], np.nan)
```

Analytical Techniques



Descriptive Statistics

Summary statistics and distributions to understand baseline market conditions.



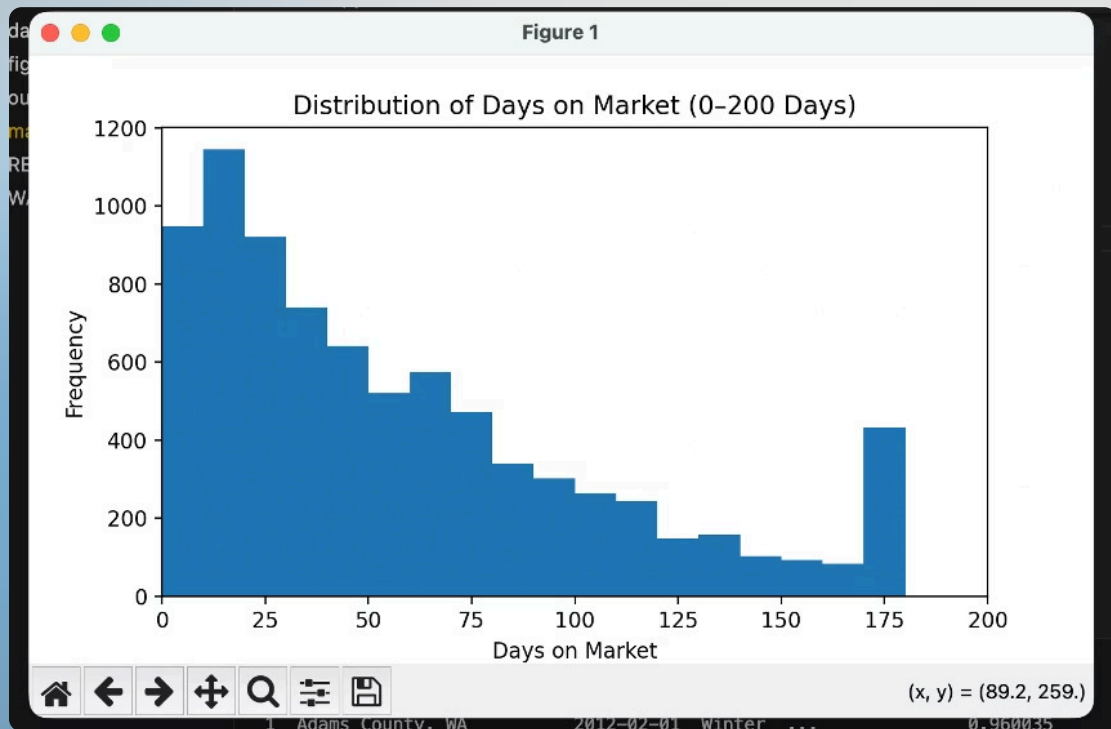
Data Visualization

Histograms, box plots, and time-series graphs using Matplotlib library.



Multiple Linear Regression

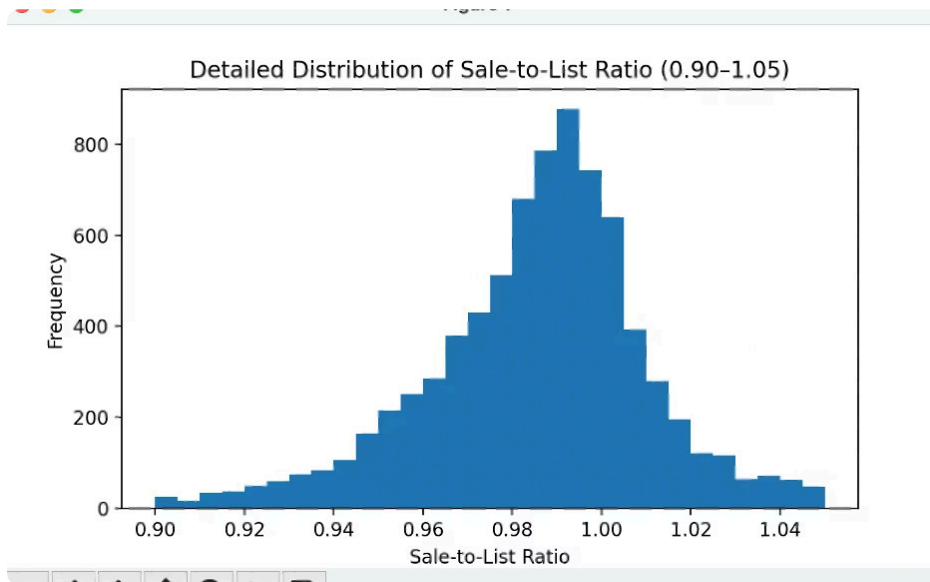
OLS models to quantify relationships and test statistical significance.



Distribution of Days on Market

The distribution shows most homes sell within 80 days, with a notable spike at 180 days—the typical listing agreement expiration period.

Sale-to-List Price Ratio Distribution



Key Finding

Distribution skews below 1.0, indicating homes typically sell for less than list price.

Interpretation

Values above 1.0 suggest strong demand; below 1.0 indicates weaker market conditions.

Seasonal Averages:

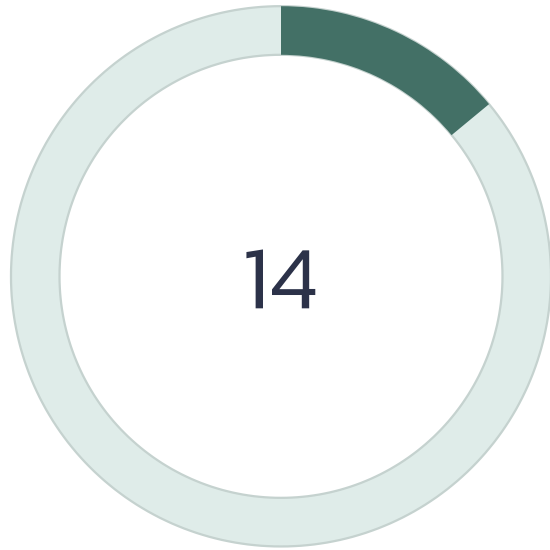
Season	Average Sale To List	Days_on_Market_capped
Fall	0.982	55.969
Spring	0.990	58.325
Summer	0.989	45.270
Winter	0.979	72.147

Seasonal Price Ratio Comparison

Spring shows the highest median sale-to-list ratio with lowest volatility. Seasonal differences are modest but consistent across observations.

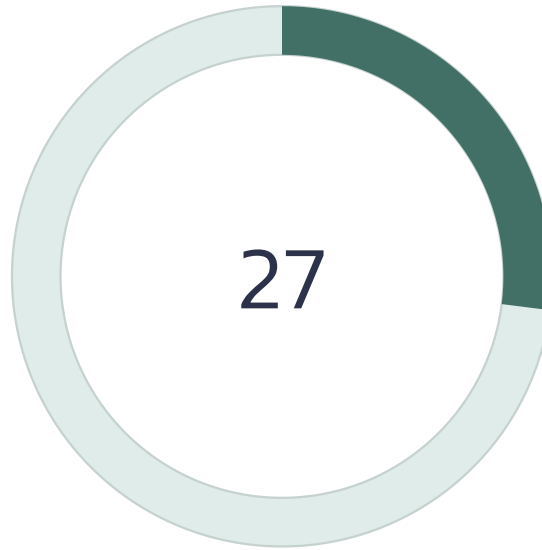
KEY FINDINGS

Seasonal Days on Market



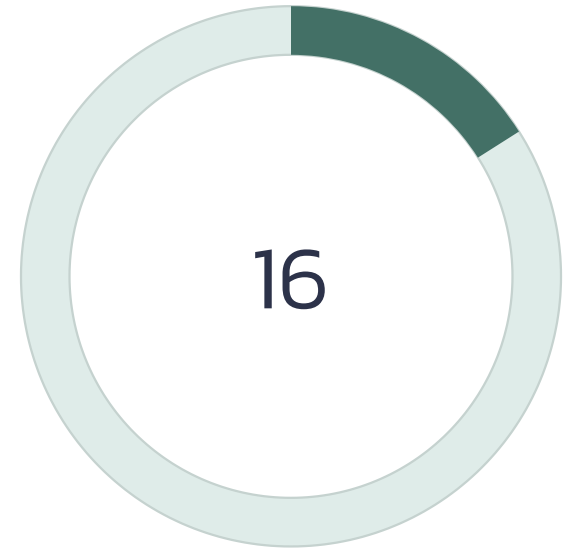
Spring

Days faster than winter ($p < 0.001$)



Summer

Days faster than winter—strongest effect



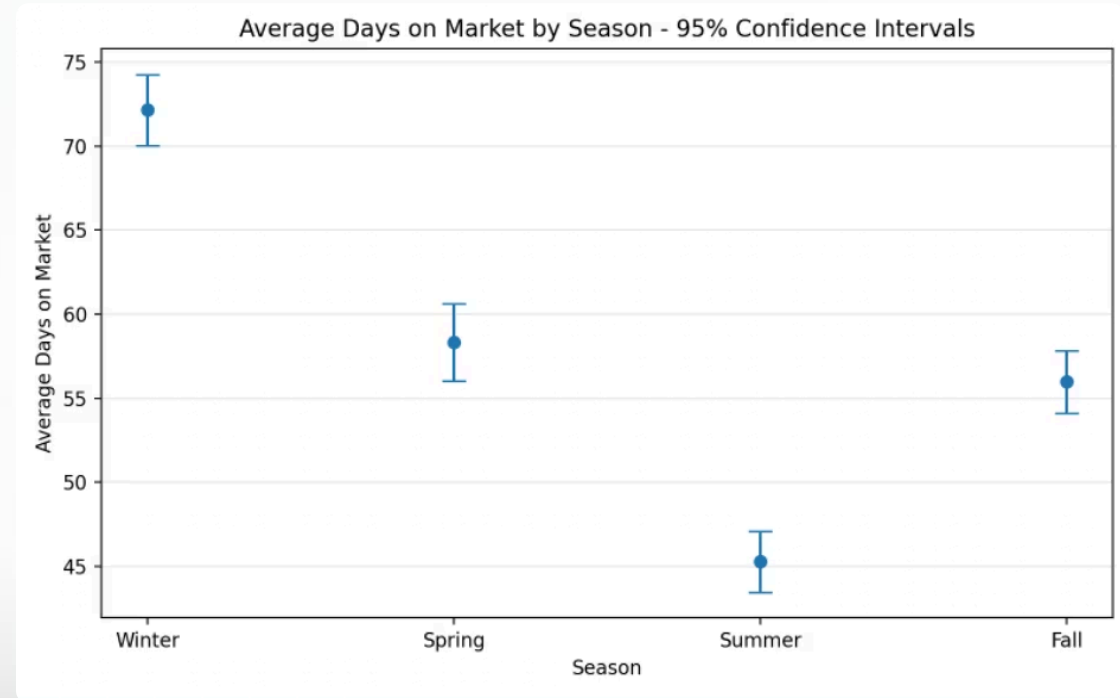
Fall

Days faster than winter

Winter is the slowest season for real estate sales, while summer provides the fastest transaction speed.

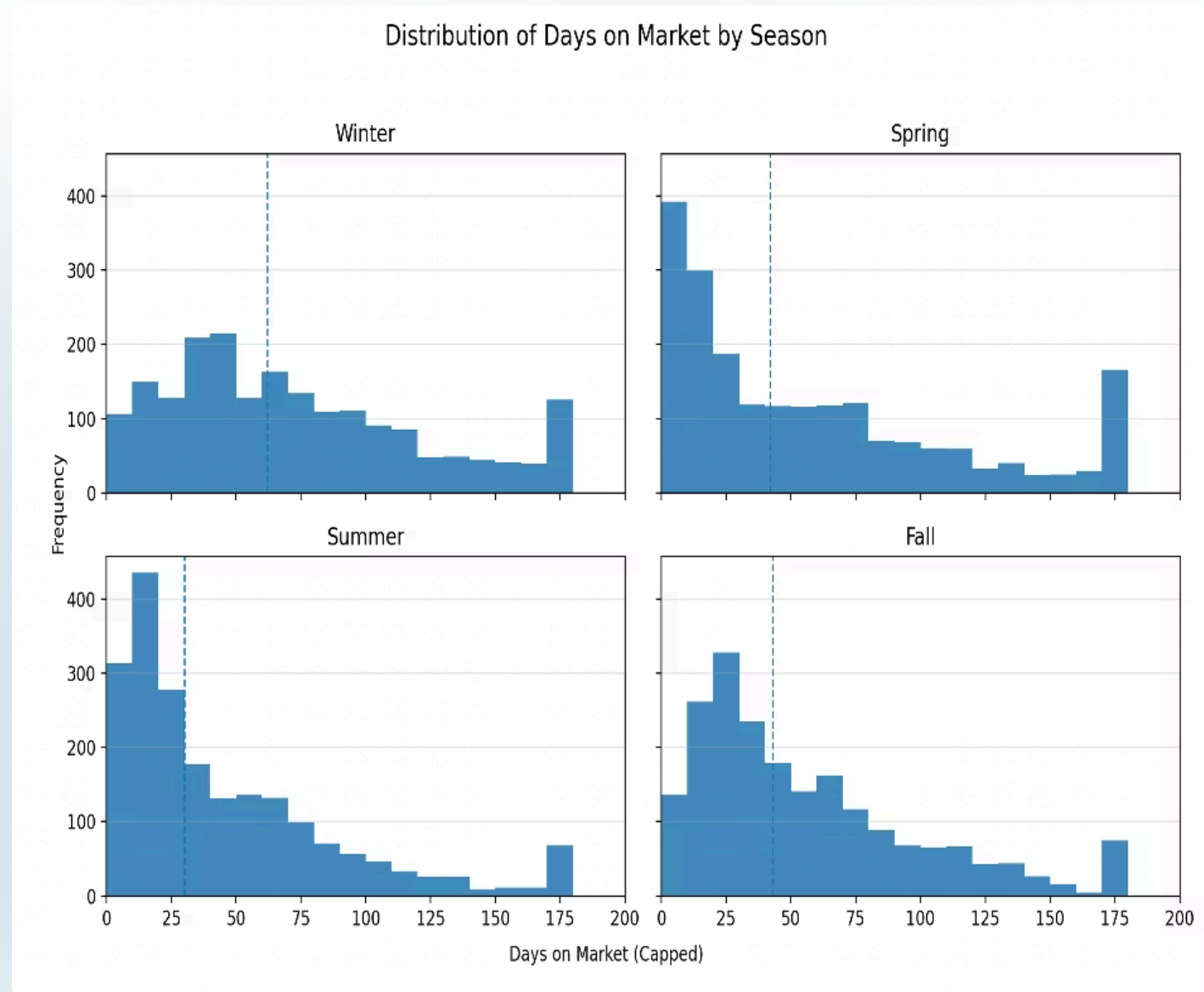
Confidence Intervals: Days on Market

95% confidence intervals reveal meaningful seasonal differences. Summer and fall show narrower bands and lower days on market, indicating stability and faster sales.



Seasonal Distribution Panels:

Summer exhibits the highest frequency of lower days-on-market values. Winter shows more even distribution, suggesting extended listing periods.



REGRESSION RESULTS

Model 1: Days on Market

0.526

Adjusted R²

52.6% of variance explained

10.8

Inventory Effect

Additional days per month of inventory

<0.001

P-Value

Highly statistically significant

Higher inventory levels reflect weaker demand, resulting in significantly longer days on market. Model far exceeds 0.10 threshold.

Model 2: Sale-to-List Price Ratio



Higher inventory weakens seller pricing outcomes. Spring and summer show statistically significant pricing advantages over winter.



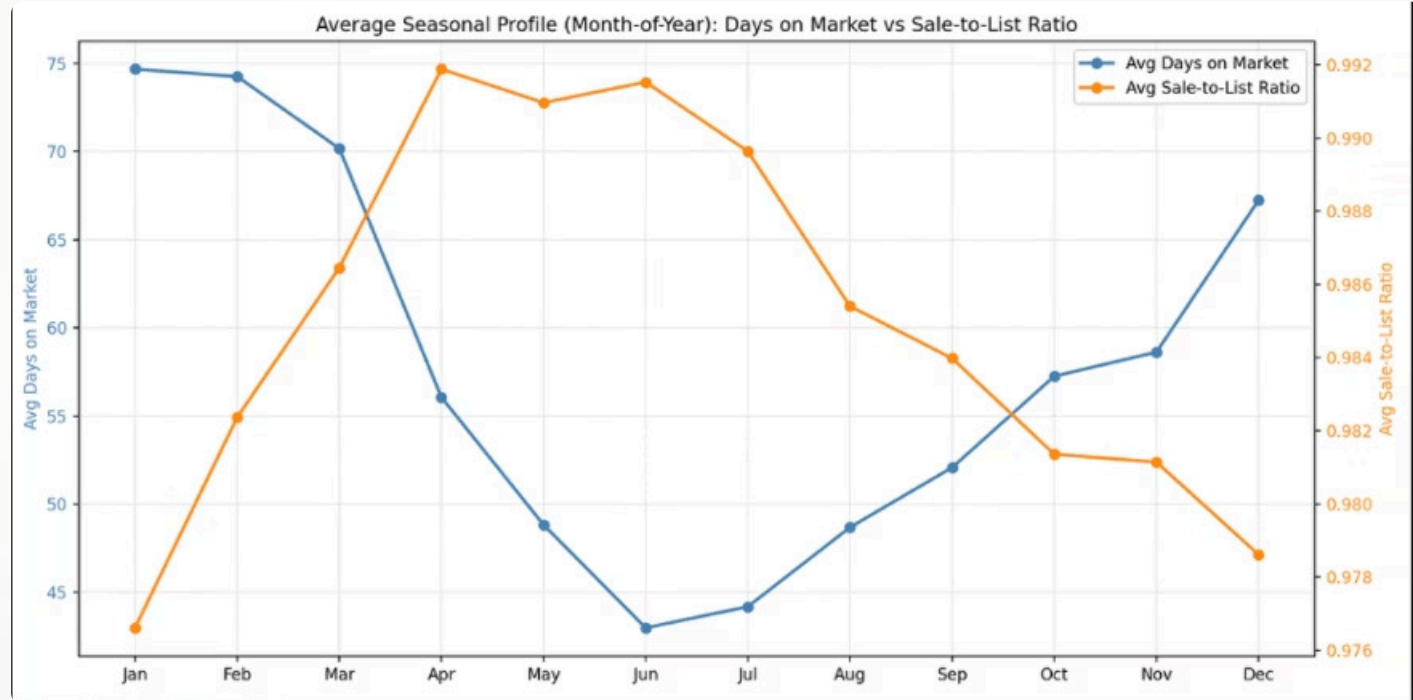
Inventory: The Strongest Driver

For every additional month of inventory, days on market increase by 10.8 days and sale-to-list ratio decreases by 0.61%.

Inventory conditions exert the most powerful influence on both transaction speed and pricing outcomes across all seasons.

Time-Series Relationship

Clear inverse relationship: elevated days on market align with lower sale-to-list ratios. Lower days on market correspond with stronger pricing performance.



PRACTICAL IMPLICATIONS

For Home Sellers



Timing Strategy

List during spring or summer for faster sales and better pricing outcomes.



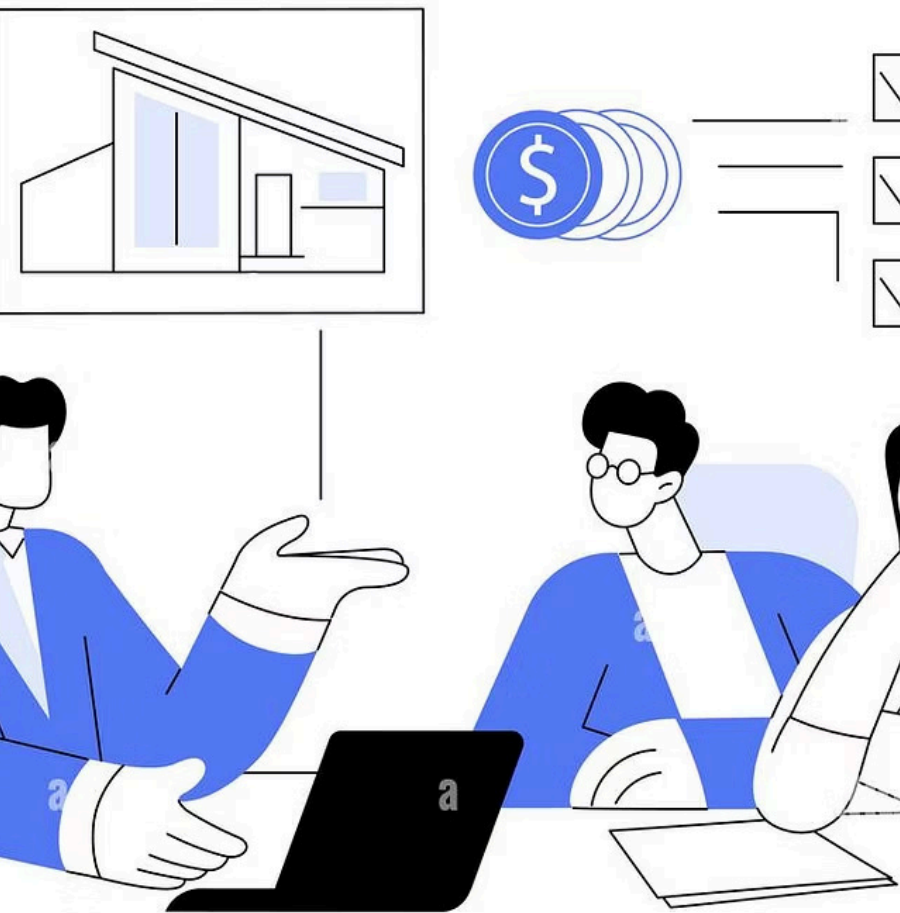
Pricing Power

1% difference can equal \$10,000 on a \$1M home—timing matters financially.



Inventory Awareness

High inventory periods require pricing concessions and longer market exposure.



For Real Estate Agents

Client Consultation

Use data to build trust, manage expectations, and guide listing strategy discussions.

Competitive Advantage

Data-driven insights differentiate agents and improve lead conversion rates.

Strategic Planning

Leverage seasonal and inventory findings to optimize timing recommendations.

Market Positioning

Provide transparent, evidence-based advice that establishes credibility.

LIMITATIONS

Study Limitations

Aggregation Bias

County-level data masks individual property characteristics like square footage, lot size, and specific location.

Omitted Variables

Model excludes macroeconomic factors such as interest rates, trade conditions, and S&P 500 performance.

Linear Assumptions

Regression assumes linear relationships, potentially oversimplifying complex market dynamics.

Future Research Directions



Property-Level Analysis

Incorporate individual home characteristics to examine interactions with seasonality and inventory.



Interest Rate Integration

Add time-specific interest rates to explain pricing outcomes and seasonal trends.



Macroeconomic Variables

Include S&P 500 averages and economic indicators to reduce unexplained variance.

CONCLUSION

Key Takeaways

Hypothesis Supported

Both models exceeded adjusted R^2 threshold of 0.10, rejecting the null hypothesis.

Inventory Dominates

Months of inventory is the strongest predictor of both speed and pricing outcomes.

Seasonality Matters

Summer provides fastest sales; spring yields best pricing. Winter shows weakest performance.

Actionable Insights

Data-driven framework empowers sellers, agents, and market participants to optimize timing strategy.

Seasonality

